



University of Central Punjab

Department of Biotechnology, Faculty of Science & Technology

Advances in Bioinformatics - BT680

Assignment

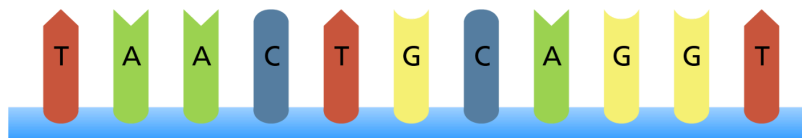
Mutation Analysis (Identification, Naming, Validation)

Instructor: Dr. Muhammad Naveed

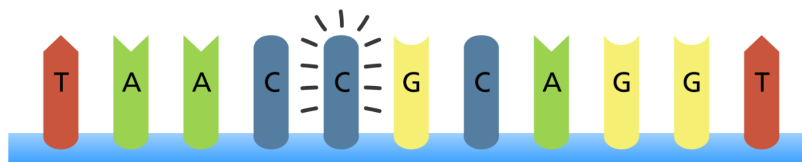
Submitted by:

Malaika Ajaz - L1S26MSBT0004

Original sequence



Point mutation



Gene: *PqqD*

1. Mutation Identification through Blastx
2. Name of Mutation: ***D39E***
3. Mutation Validation:

Mutation Validation Tools	Results
Polyphen 2	Benign
MU pro	Decreased
I Mutant	Decrease Stability
PhD-SNP	Deleterious
SIFT	Tolerate (no effect on structure)

Results

The D39E mutation identified in the PqqD gene was analyzed using multiple validation tools to evaluate its impact on protein structure and function. Since most tools predicted negative effects, the D39E mutation is likely to affect the stability and functional activity of the PqqD protein.